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 Scrum.org



Understanding the Professional Scrum Developer training and certification program

Richard Hundhausen
Accentient Corporation

Ken Schwaber
Scrum.org



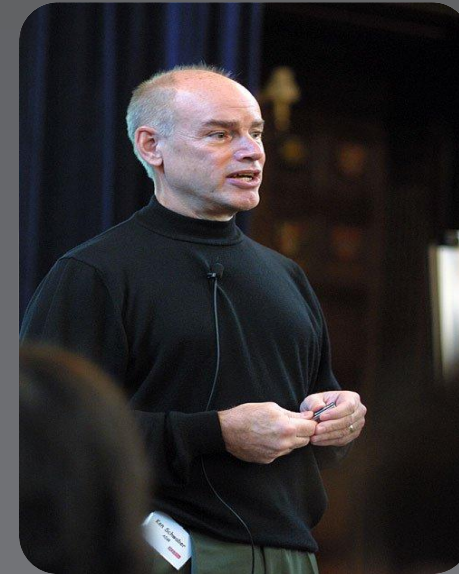
Who Are These Guys!?

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Richard Hundhausen



Ken Schwaber



Manifesto for Agile Software Development

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We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

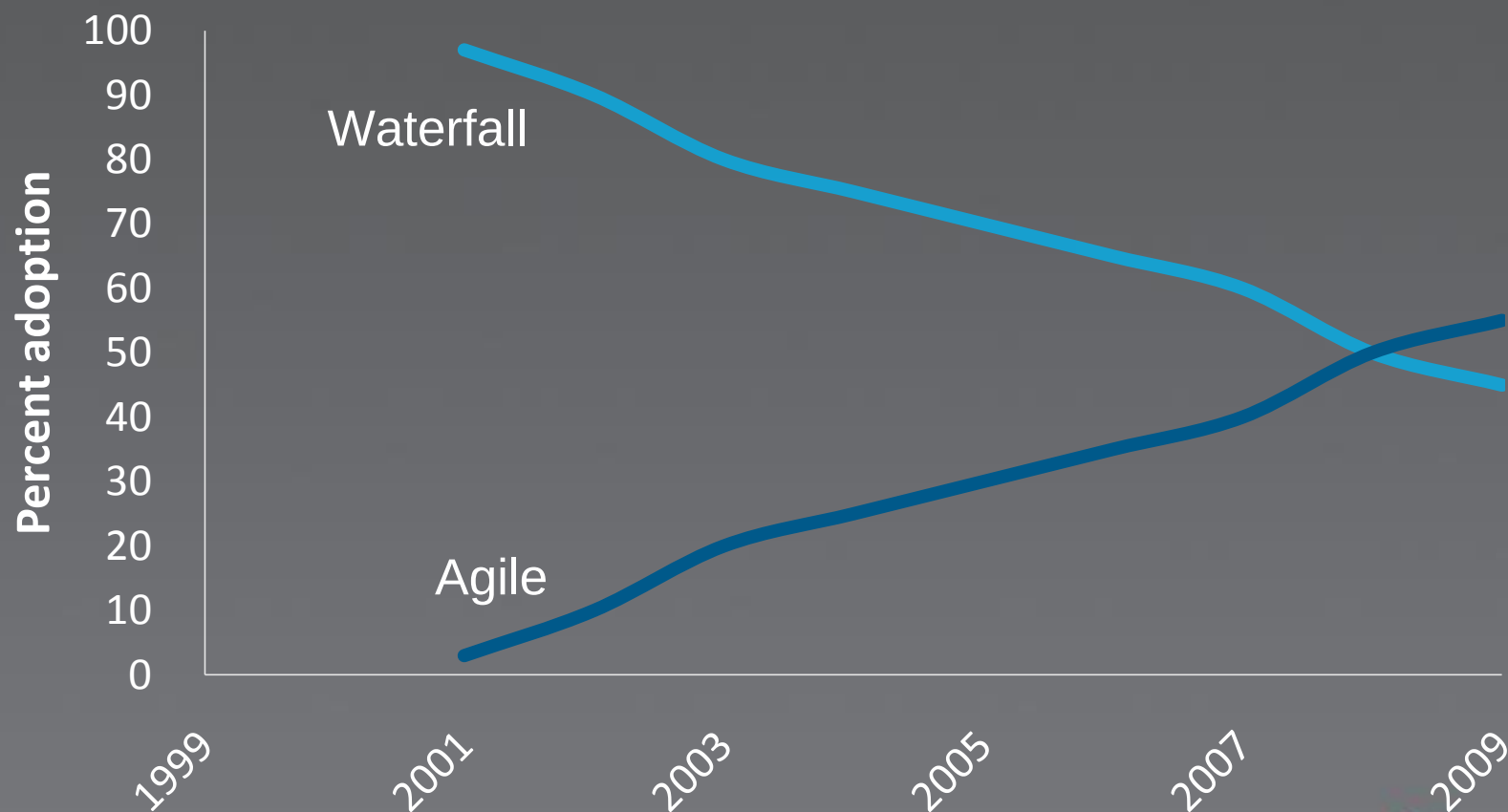
Kent Beck
Mike Beedle
Arie van Bennekum
Alistair Cockburn
Ward Cunningham
Martin Fowler

James Grenning
Jim Highsmith
Andrew Hunt
Ron Jeffries
Jon Kern
Brian Marick

Robert C. Martin
Steve Mellor
Ken Schwaber
Jeff Sutherland
Dave Thomas

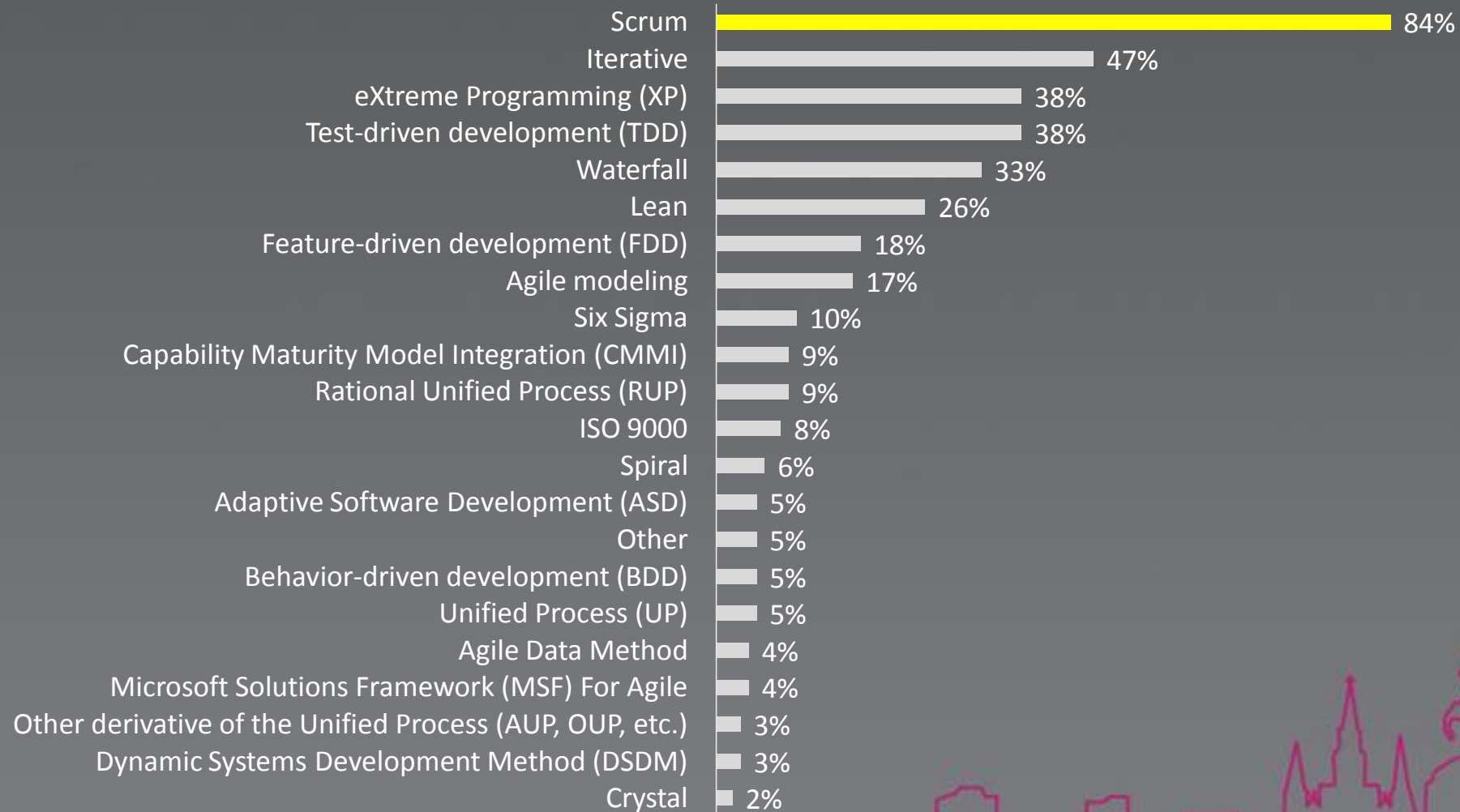
Agile Overtook Waterfall in 2008

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Most Development Organizations Using Agile Use Scrum 6

“Check all techniques or methodologies that you currently use, wholly or partially.”



Base: 241 technology industry professionals in a variety of roles, including but not limited to development

Scrum Reduces the Planning Horizon to One Sprint, Increasing Predictability & Reducing Risk

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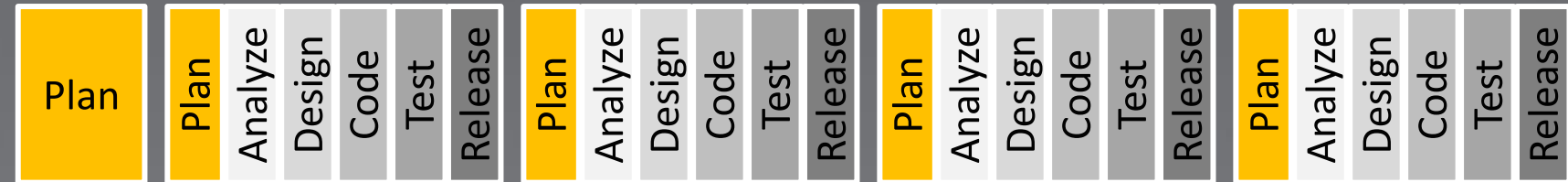
Waterfall

Plan for the entire project up-front



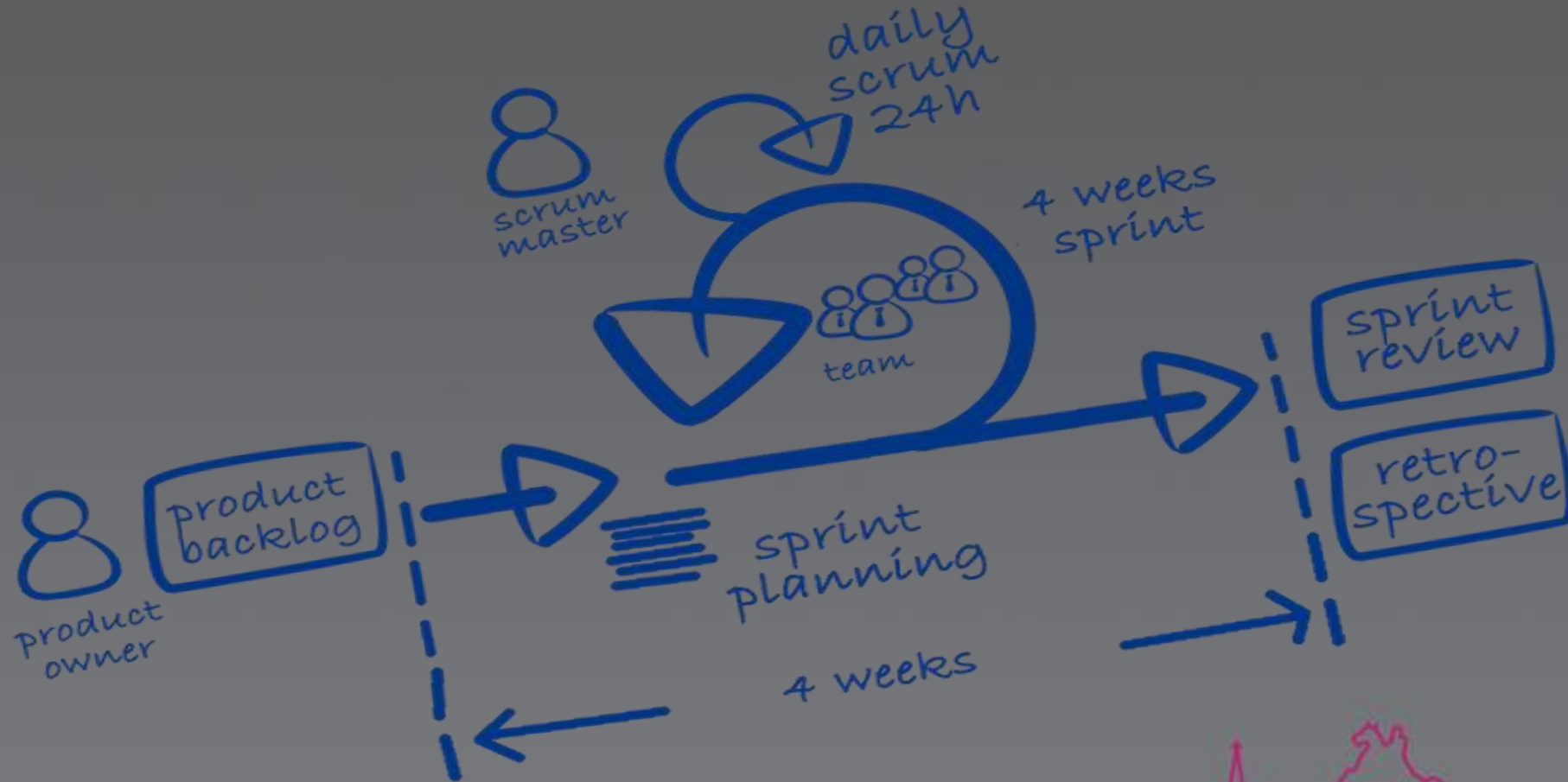
Scrum

Plan a little for the entire project and then a little for each Sprint



The Scrum Framework Is Simple, Full of Holes

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You're Welcome

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But Agile Alone Can't Make Developers Great

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In 2002, in response to the Agile Manifesto, Barry Boehm was quoted as saying,

“Agile is an excellent approach is you have a small team of highly skilled developers managing themselves in a co-located workplace with great engineering tools and practices.”

Teams like these will produce great products using Agile.

It also turns out that you can also use Agile with a large team of terrible developers who are dispersed all over the globe, who are using lousy tools and practices.

Teams like these will produce crap.

*The point isn't whether they produce great products or crap.
The point is that with Agile, the problem is transparent.
Then the question is, what are you going to do about it?*

Martin Fowler Calls Scrum “Flaccid”

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“There's a mess I've heard about with quite a few projects recently. It works out like this:

- They want to use an agile process, and pick Scrum
- They adopt the Scrum practices, and maybe even the principles
- After a while progress is slow because the code base is a mess

What's happened is (people using Scrum) haven't paid enough attention to the internal quality of their software (...) I've mentioned Scrum because when we see this problem, Scrum seems to be particularly common as the nominative process the team is following (...) because Scrum is process that's centered on project management techniques and deliberately omits any technical practices.

I'm sure that the many **Flaccid Scrum** projects being run will harm Scrum's reputation, and probably the broader agile reputation as well.”

– Martin Fowler, January 2009



Even Ken and Jeff Are Worried!

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“By early 2009, (...) more organizations were using Agile processes than waterfall processes (...) However, less than 50% of those using Scrum were developing in incremental iterations, which are the heartbeat of Scrum. (...) One of the biggest challenges of using Scrum has always been the steep learning curve for the developers on the Scrum team.”

– Jeff Sutherland and Ken Schwaber, March 2010

Exercise #1: Five Minutes with Three People Near You

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The Situation:

- You are a developer at xyz.co, building advanced life-critical products.
- Your Scrum team is one of seven teams working on a new release of one of the products.
- Your team is going to select requirements (product backlog) to turn into something that is done (no more work remains, potentially shippable) within a two-week iteration.
- Each team has all the skills to fully develop the requirements into a “done increment.”

The Assignment:

- What work would you have to do to turn the requirements into a “done” increment?
- If you were developing a “done”, potentially shippable increment, what would your definition of “done” be? Would it include, for example, refactoring? What else?

Exercise #2: Five Minutes with Three Different People

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The Assignment:

Did your definition of “done” include the following? If not, why not?

Code reviews

Refactoring

Integration with other team's work

Integration testing

Regression testing

Performance testing

Stability testing

Immunological response testing

Release notes

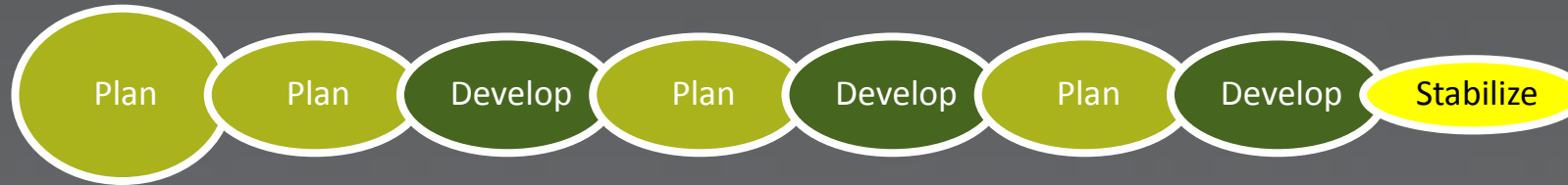
Internationalization

User acceptance testing

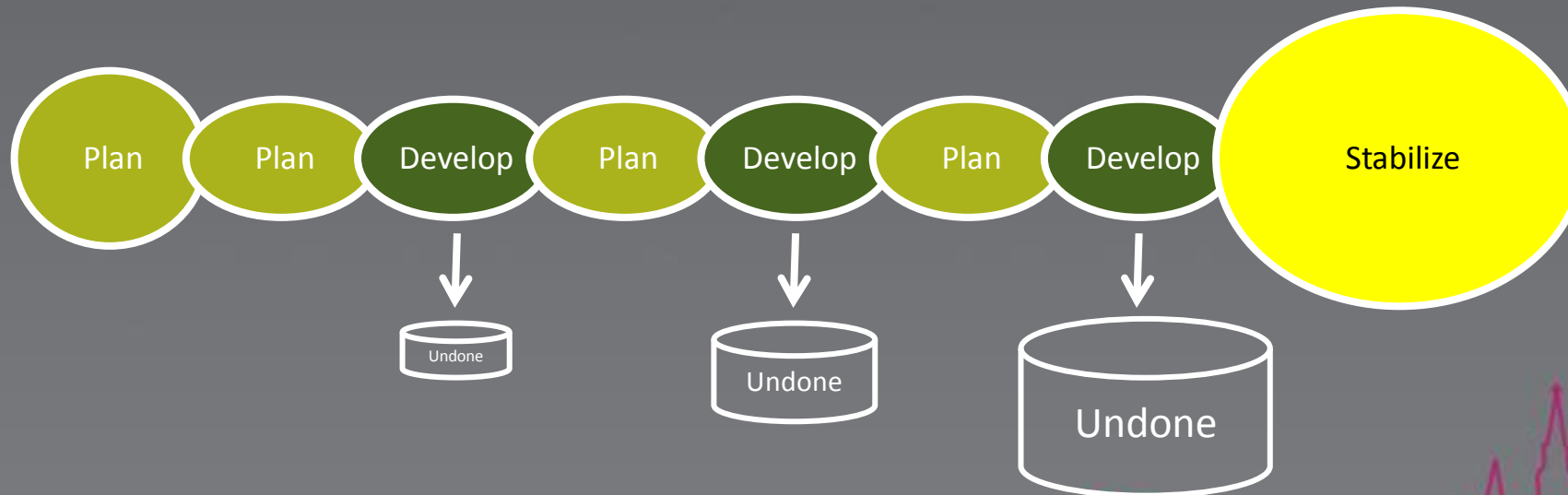
“Stabilization”: When You Do All The “Undone” Work

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Scrum project with complete, integrated “done”:



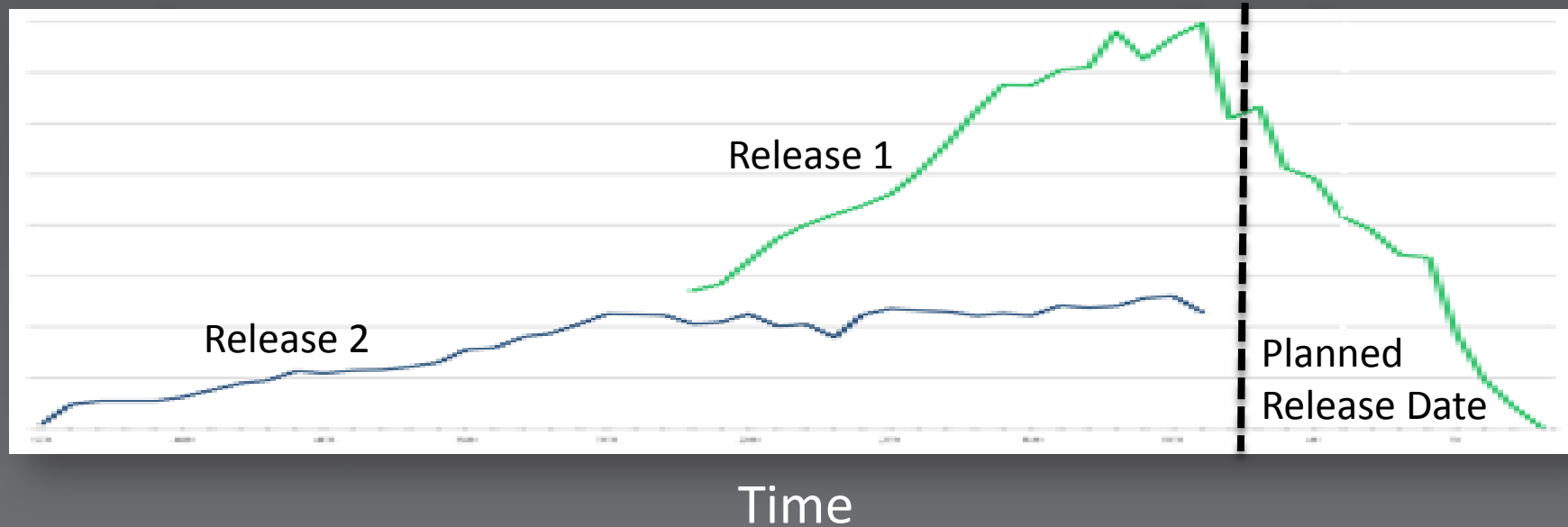
Scrum project with incomplete or variable “done”:



Better Engineering Practices Support Predictability

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Defects



- Release 1: Teams produced “done” increments each Sprint, but they were not integrated or integration tested until “code complete.”
- Release 2: Teams produced an increment of integrated, integration-tested code every Sprint.

A case study featuring 120 people divided amongst 18 Scrum teams



Martin Fowler's Call to Action

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“The Scrum community needs to redouble its efforts to ensure that people understand the importance of strong technical practices. Certainly any kind of project review should include examining what kinds of technical practices are present. If you're involved or connected to such a project, make a fuss if the technical side is being neglected.”

Martin Fowler, January 29, 2009





Who, me?!

What are you going to do about it?

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**“But I’m a developer,
not a manager!”**

- Learn about Scrum
- Get hands-on Scrum training
- Measure your understanding
- Pass it on



**“But I’m a manager,
not a developer!”**

- Learn about Scrum
- Get your teams hands-on Scrum training
- Measure their understanding
- Set a bar

Improving the Profession of Software Development

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The Purpose of Scrum.org

Maintain Scrum



Create and maintain bodies of knowledge on which to train and assess people for the betterment of software development.

Offer Training



Work with experts in various technologies and fields to create and deliver courses that meet the needs of the software development profession. Continuously monitor quality.

Assess Knowledge



Offer rigorous assessments to allow people to evaluate their abilities so they are able to improve.

Introducing the Professional Scrum Developer Program

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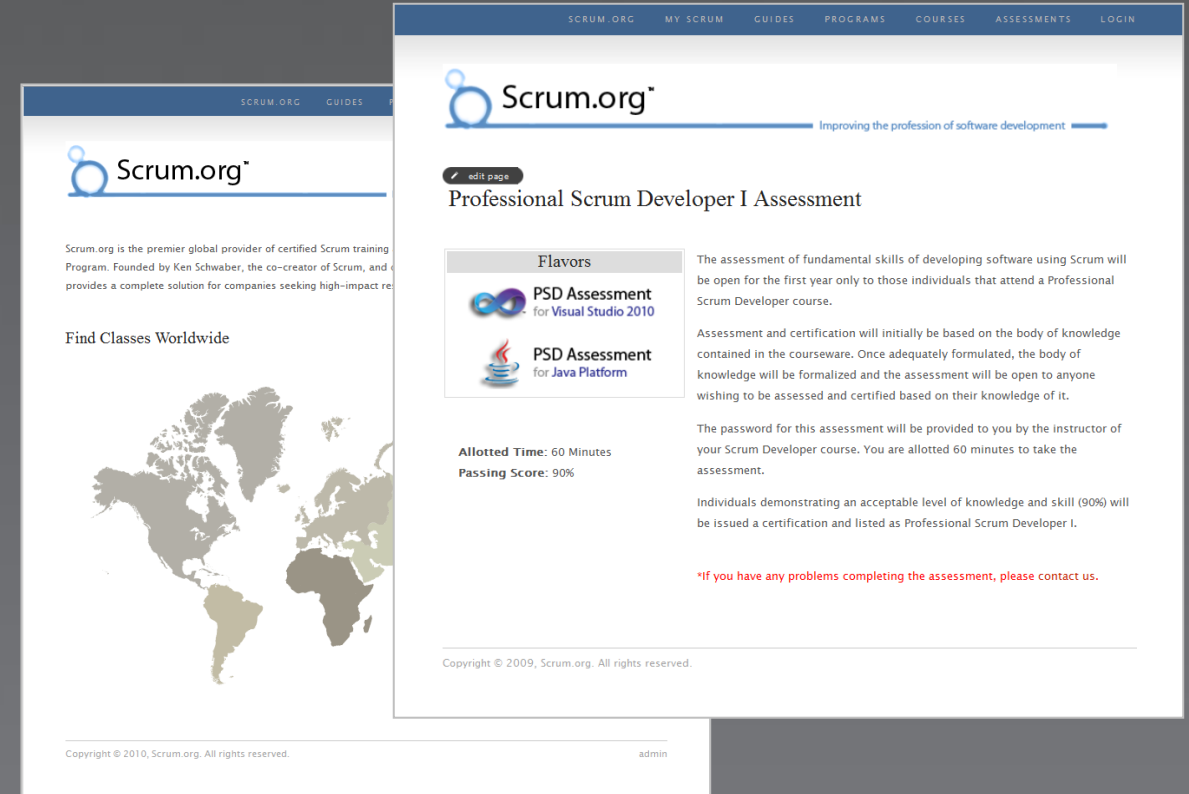
- An innovative program for developers from Microsoft and the founders of Scrum
- Including a training course, assessment, and certification
- Visit Scrum.org (<http://www.scrum.org>) or MSDN (<http://bit.ly/dppXd0>) for more details



Four Pillars of the Professional Scrum Developer²³ Program

Learn how to:

1. How to work together as a cross-functional, self-organizing team,
2. Using modern engineering practices,
3. On a modern technology stack, in a modern development environment,
4. To build a “done” increment within an iteration.



The Official Microsoft Training for VS 2010

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Teaches VS2010 and best practices, all within the Scrum framework

Goes beyond traditional “tool-only” courses

All the content you need in an efficient and effective 5-day format

The most efficient way to ensure retention of necessary material

Delivered publicly or privately to meet varying needs of customers

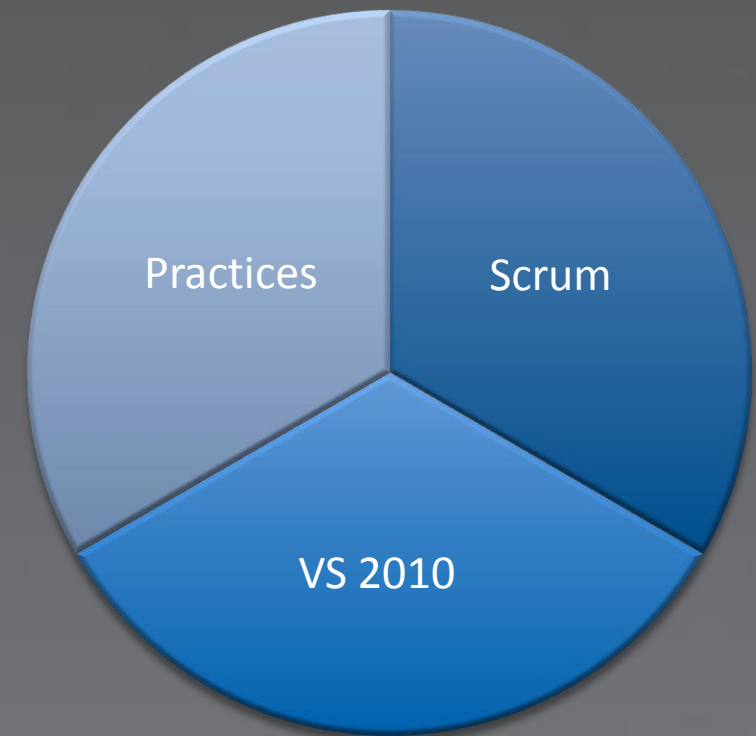
Reduces both cost and disruption to developers

Being delivered with the **NEW** Scrum Process Template

New Microsoft-developed Scrum Process Template for VS2010 is the most faithful to the Scrum process; available on MSDN in June 2010.

A certification with teeth that differentiates skilled developers

Must actively participate in all 5 days of the course and pass a rigorous assessment with a score of **at least 90%** to receive certification.



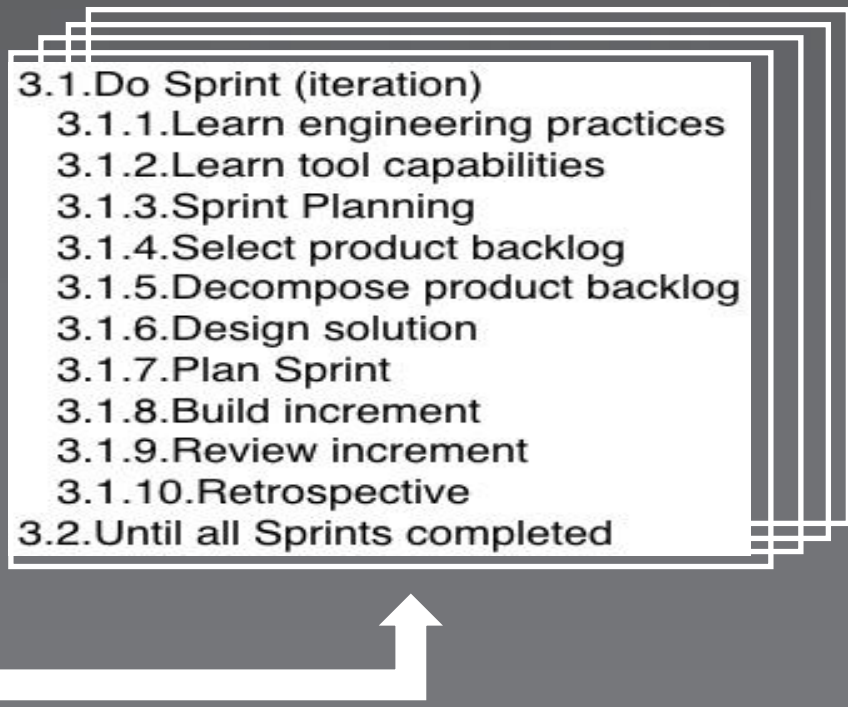
Course Structure Mirrors the Structure of Scrum

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Teams iteratively build increments while learning more Scrum, teamwork, engineering techniques, and tooling each Sprint. They are allowed to fail and learn.

Structure of initial Sprint:

- Start
- Initiation
- Form team
- Course overview
- Case study overview
- IDE overview
- Scrum overview
- Develop “product”
- Retrospective



3.1. Do Sprint (iteration)
3.1.1. Learn engineering practices
3.1.2. Learn tool capabilities
3.1.3. Sprint Planning
3.1.4. Select product backlog
3.1.5. Decompose product backlog
3.1.6. Design solution
3.1.7. Plan Sprint
3.1.8. Build increment
3.1.9. Review increment
3.1.10. Retrospective
3.2. Until all Sprints completed

Each Sprint introduces new:

- Engineering practices
- IDE and technology features
- Sample product backlog

Sample Question: Development (1 of 5)

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Question 7 of 88

What is the primary purpose of refactoring?

- ☐ **a)** Removing all bugs that were found during regression tests.
- ☐ **b)** Ensuring that all factors are constantly aligned.
- ☐ **c)** Making sure that the code is readable and maintainable.
- ☐ **d)** Creating better technical documentation.

Sample Question: Development (2 of 5)

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Question 6 of 88

You encounter a bug. You are filling out a bug report. What are attributes of a good bug report?

- ☐ **a)** Clear title and proper grammar in report.
- ☐ **b)** One bug per report.
- ☐ **c)** Simple and repeatable reproduction steps.
- ☐ **d)** Expected results and observed results.
- ☐ **e)** Build/version where bug was found.
- ☐ **f)** Screenshots or other pictures of bug in action.

Sample Question: Development (3 of 5)

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Question 2 of 151

Architectural layering is an important consideration because:

- ☐ **a)** You increase the portability of your system by reducing the coupling between classes.
- ☐ **b)** It reduces the chance of circular references.
- ☐ **c)** It is a class separation of concerns.
- ☐ **d)** It is a best practice and therefore easy to understand for all team members.
- ☐ **e)** All of the above.

Sample Question: .NET (4 of 5)

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Question 10 of 88

Which of the following mechanisms can be used to prevent bad architectural layering (choose 2)?

- ☐ **a)** Creating a Use Case diagram in VS2010.
- ☐ **b)** Make sure that all design models are always up-to-date.
- ☐ **c)** Creating and validating a layer diagram in VS2010.
- ☐ **d)** Design the baseline architecture before the beginning of Sprint 1.
- ☐ **e)** Reinforce the intended architecture during the development and maintenance of your code by including validation with your check-in and build operations.

Sample Question: .NET (5 of 5)

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Question 9 of 151

When is the best time to use gated Check-ins?

- ☐ **a)** When you have an inexperienced Product Owner.
- ☐ **b)** When the Team is inexperienced at using Version Control and automated Builds.
- ☐ **c)** When the team is experienced with TDD and proficiently uses Version Control and automated Builds.
- ☐ **d)** When you have an inexperienced ScrumMaster.

A Long, Hard, Worthwhile Climb

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Questions?

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Resources



Sessions On-Demand & Community

www.microsoft.com/teched



Microsoft Certification & Training Resources

www.microsoft.com/learning



Resources for IT Professionals

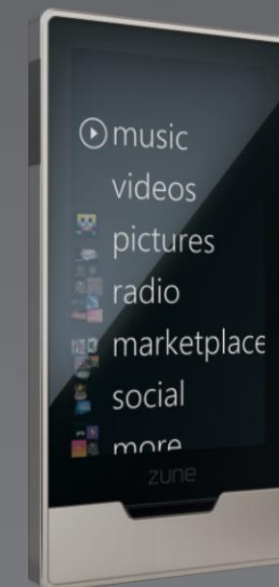
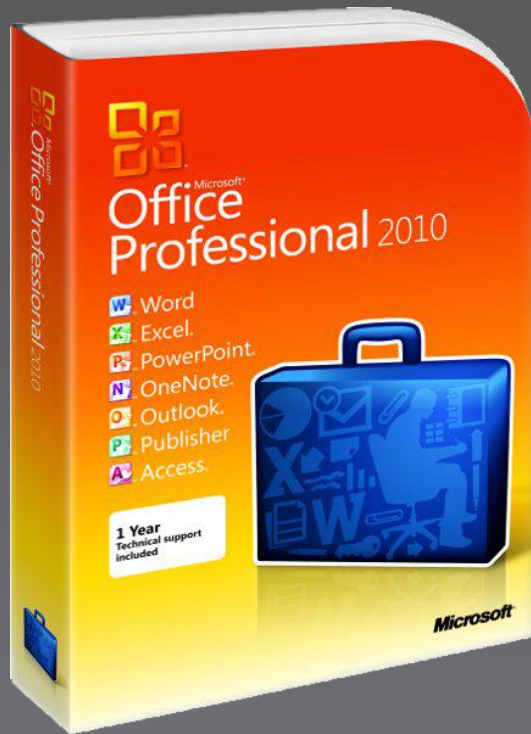
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Resources for Developers

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CommNet and
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